XINYUAN (ELLA) LI

San Diego, CA · +1 612 458 9922 · ell024@ucsd.edu · linkedin.com/in/ellali86 · github.com/EllaLi86

SUMMARY

Data Science and Business Economics student at UC San Diego (GPA 3.99) who turns financial and operational data into decisions: built a real-estate valuation model (11.3% MAPE) at a proptech startup, shipped an AI tutoring product adopted by 60% of 400+ students, and currently build Tableau dashboards as a Data Analysis Intern. Seeking Summer 2027 business and data analyst internships.

EDUCATION

University of California, San Diego | *Halıcıoğlu Data Science Institute* Expected June 2028
B.S. Data Science and Business Economics · GPA 3.99/4.0 · Provost Honors, all eligible quarters
Relevant Coursework: Machine Learning, Introduction to Data Visualization, Data Structures & Algorithms for Data Science, Theoretical Foundations of Data Science I & II, Practice of Data Science, Intro to Data Management, Applied Statistical Data Analysis, Econometrics A/B/C

RESEARCH

Student Researcher, AI in Education | *Data Science Teaching and Learning Lab, UC San Diego* Oct 2025 – Present
- Built and shipped a full-stack JupyterLab extension (React/TypeScript, Python/Tornado) embedding an AI tutor in student notebooks, adopted by 60% of 400+ students across 2 quarters of DSC 10, UC San Diego's introductory data science course, and logging ~88,000 tutoring sessions.
- Designed a context-aware retrieval layer that sanitizes live notebook state — active cell, markdown instructions, and outputs — and surfaces relevant lecture examples via TF-IDF retrieval with Gemini reranking, reducing generic/off-topic tutor responses by grounding answers in course-specific material.
- Analyzed session logs to identify usage patterns (e.g., peak usage before deadline) and used them to prioritize new features: exam mode, topic-based practice retrieval, and Socratic prompting.

Student Researcher, Economics | *Department of Economics, UC San Diego* Jun 2026 – Present
- Estimated how UC faculty summer teaching supply responded to the 2019 removal of the $10,000-per-course cap on summer pay (set at 8.5% of base salary), applying regression discontinuity and difference-in-differences to 34,585 administrative course records.
- Built the full cleaning-to-estimation pipeline in Stata and ran robustness checks across bandwidths and specifications; results feed into a working paper.

WORK EXPERIENCE

Data Analysis Intern | *Y Montessori* Oct 2026 – Present
- Contributing to the Montopia Data Lens monthly report, collecting and analyzing education research datasets to surface trends for educators and partner schools.
- Building public-facing Tableau visualizations and designing dashboard layouts in Figma, turning education data into clear, accessible insights.

Data Science Tutor | *Halıcıoğlu Data Science Institute, UC San Diego* Sep 2025 – Present
- Owned an automated grade-reporting pipeline (Python, bash, Gradescope API) that computes final scores across labs, homework, quizzes, exams, and participation — handling slip days, drops, and curving — then pushes personalized reports to more than 200 students weekly, replacing an error-prone manual process and ensuring consistent, reproducible grading.
- Maintained the course website and GitHub repository, releasing weekly labs, midterm and final projects on schedule across 3 quarters.
- Led discussion sections, pod meetings, office hours, and one-on-one exam reviews for more than 200 students per quarter on regression, permutation testing, and Python in Jupyter.

Data Scientist Intern | *IDX Exchange* Jun 2025 – Aug 2025
- Built an automated valuation model (AVM) on 125,542 real-estate transactions (CoreLogic Trestle API) to estimate property market value, the core input behind listing-price recommendations on the platform.
- Engineered 50+ features (location, living area, property age, sale timing) and used exploratory analysis to isolate the price drivers that shaped the final model architecture.
- Trained and tuned linear regression, random forest, and XGBoost valuation models, cutting a 20.9% MAPE baseline through iterative tuning (log-transforming price, pruning low-value features) to a final 11.3% MAPE (R^2 0.82).
- Set reproducibility standards (environment pinning, seeded splits, model versioning) adopted by the team; presented results weekly to senior data scientists.

TECHNICAL SKILLS

Languages: Python (pandas, NumPy, scikit-learn, XGBoost), R, SQL, Stata, TypeScript/JavaScript, Java, HTML/CSS
Machine Learning and Statistics: Regression modeling, feature engineering, A/B testing, permutation testing, causal inference (difference-in-differences, regression discontinuity), exploratory data analysis
AI and LLMs: Gemini API, prompt engineering, retrieval-augmented generation (TF-IDF retrieval, LLM reranking)
Tools: Tableau, Excel (advanced), Git/GitHub, JupyterLab, PostgreSQL, React, Streamlit, Matplotlib, Seaborn, Cursor
Spoken Languages: English (fluent), Mandarin (native)

LEADERSHIP, HONORS, AND AWARDS

Orientation Leader | *Warren College, UC San Diego* Apr 2025 – Oct 2026
- Presented campus resources to more than 300 incoming students and synthesized post-event survey data from respondents to reshape the following year's programming.

Events Board Member | *Warren College Student Council, UC San Diego* Sep 2024 – Sep 2025
- Managed and tracked a $27,581 budget across 20 campus events for 500+ attendees, allocating funds by event priority and reconciling spend against plan.

Fourth Place, Programming Competition*, Women in Computing, UC San Diego* 2025